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StuntGraph: An Integrated Graph-Based Spatio-Temporal Framework for Analyzing Socioeconomic Determinants of Stunting Prevalence Across Indonesian Provinces Through Machine Learning, Spatial Network Analysis, and Community Detection

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Abstract: This study presents StuntGraph, a graph-based spatio-temporal framework for identifying socioeconomic determinants of stunting and priority intervention regions across 34 Indonesian provinces (2021–2024). Using data from BPS and Kemenkes RI, the framework integrates PCA, Multiple Linear Regression (MLR), Spatial Network Construction, K-Means Clustering, and Louvain Community Detection. PCA identifies education, sanitation, and clean water access as the main contributors to stunting variation, while regression confirms poverty (β = 0.2635, p = 0.039) and sanitation (β = -0.2013, p = 0.015) as significant predictors. For predictive benchmarking, Random Forest and XGBoost were evaluated alongside MLR. MLR achieved the best performance (RMSE = 5.14, R^2 = 0.345), outperforming Random Forest (R^2 = -0.098) and XGBoost (R^2 = -1.370). K-Means clustering (k = 3, Silhouette Score = 0.2025) classified provinces into low, medium, and high-risk groups, with high-risk provinces concentrated in Eastern Indonesia. Louvain community detection identified 13 spatial communities, supporting the west-to-east stunting gradient. Combining clustering and graph-based analysis provides a basis for spatially differentiated stunting interventions.

Keyword: *stunting prevalence, spatio-temporal analysis, K-Means clustering, Louvain community detection, PCA, spatial network, Indonesian provinces*

CHAPTER 1 Introduction

A. Background

Stunting is a condition of chronic malnutrition in children under five, characterized by a height significantly below the standard for their age. Beyond physical growth, stunting has serious long-term consequences including impaired cognitive development, poor academic performance, and reduced economic productivity in adulthood. In Indonesia, stunting remains a critical public health challenge, despite a declining trend from 27.7% in 2019 to 19.8% in 2024, the distribution across provinces remains highly uneven, with several regions still reporting rates well above the national average (Kemenkes RI, 2024). The Indonesian government has set a national target of reducing stunting prevalence to 14% by 2024 under the National Medium-Term Development Plan (RPJMN 2021-2024).

Stunting is deeply intertwined with socioeconomic conditions such as poverty, limited access to clean water and sanitation, low maternal education, and inadequate public health services. Research also confirms a direct long-term causality between stunting prevalence, poverty, and economic growth across Indonesian provinces, where stunting can result in annual GDP losses of up to 3.99% (Yudhistira et al., 2021). Moreover, spatio-temporal analysis has shown that stunting risk varies significantly across provinces, with regions such as Nusa Tenggara Timur and Sulawesi Barat consistently reporting the highest relative risk (Rahmawati et al., 2024).

Despite the growing literature on stunting in Indonesia, most studies have focused on national-level or single-province analyses, lacking an integrated spatial perspective across provinces. To address this gap, this research introduces StuntGraph: a graph-based spatio-temporal framework combining PCA, Multiple Linear Regression, Machine Learning, Spatial Network Construction, and Community Detection. Using data from 34 provinces over 2020–2024, this study aims to identify dominant socioeconomic drivers of stunting and detect priority regions for government intervention.

B. Problem Statement

Based on the background above, the research question are:

1. What socioeconomic factors most significantly influence stunting prevalence across Indonesian provinces?
2. What spatial network patterns exist among provinces based on stunting characteristics?
3. Which provinces share similar stunting risk profiles and should be prioritized for intervention?

C. Research Objectives

This research aims to:

1. Identify dominant socioeconomic factors affecting stunting prevalence using EDA, PCA, and MLR, while benchmarking MLR, Random Forest, and XGBoost for prediction accuracy.
2. Construct and visualize a spatial network model of inter-provincial relationships based on geographic adjacency and stunting characteristics.
3. Detect and evaluate province clusters with similar stunting risk profiles using Louvain Community Detection and K-Means Clustering to support intervention prioritization.

CHAPTER 2 Research Methodology

A. Dataset Description

This research uses secondary data collected from two official government sources covering 34 Indonesian provinces over the period 2021-2024. Socioeconomic variables including poverty rate, unemployment rate, sanitation access, clean water access, average years of education, population density, and GDRP per capita were obtained from Badan Pusat Statistik (bps.go.id). Stunting prevalence data per province was retrieved from the Kementerian Kesehatan Republik Indonesia (kemenkes.go.id). Additionally, Indonesia’s province boundary shapefile for spatial network construction was sourced from (indonesia-geospasial.com). The final integrated dataset forms a spatio-temporal panel of 34provinces x 5 years, resulting in 136 observations.

B. Research Variables

The research's variables include the prevalence of stunting as a dependent variable and a number of independent variables that are derived from socioeconomic indicators, including population density, GRDP per capita, average years of schooling, unemployment rate, access to clean water, proper sanitation, and population density. Additionally, in this study, spatiotemporal analysis and spatial network analysis are supported by supporting variables like province and year.

Table 1. Independent Variable

Symbol	Variable Name
X1	Poverty Rate
X2	Unemployment Rate

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X3	Proper Sanitation Access
X4	Clean Water Access
X5	Average Years of Schooling
X6	Population Density
X7	GRDP per Capita

Table 2. Identifier Variable

Symbol	Variable Name
ID1	Province
ID2	Year

Table 3. Target Variable

Symbol	Variable Name
Y	Stunting Prevalence

C. Data Preprocessing

In order to create a clean, consistent, and ready to use dataset for additional analysis, data preparation is an essential step in this research. An adjustment procedure is required initially since the data from Republic of Indonesia's Ministry of Health and the Central Statistics Agency have different formats and structures. At this point, duplicate data is eliminated, provincial name spelling errors are fixed, and data type uniformity across all research variables is ensured.

Following the data cleaning, a spatiotemporal panel dataset was created by integrating all datasets according to the province and year variables. Data transformation was the next stage, which involved integrating geographic data using shapefiles or GeoJSON files for Indonesian provinces, standardising numerical forms, and modifying data structures. Outlier checks were also performed using boxplot visualizations and descriptive statistical analysis to detect extreme values that could potentially affect the analysis results. Outliers deemed irrelevant or resulting from input errors were then adjusted to maintain dataset quality.

The preprocessing stage also includes data normalization and feature scaling, as each variable has different units and value ranges. The scaling process is performed using methods such as StandardScaler or Min-Max Scaling to ensure each variable contributes equally to PCA, clustering, and graph modeling. Additionally, spatial data preparation is performed by representing each province as a node and the relationships between provinces as edges based on geographical proximity or administrative boundaries.

D. Principal Component Analysis (PCA)

By converting the original variables into a set of uncorrelated principal components arranged linearly, Principal Component Analysis (PCA) is a dimensionality reduction technique frequently used in multivariate analysis to find patterns and relationships among variables in complex datasets (Gewers et al., 2021). The first principal component in the dataset captures the greatest variation. Each principal component is built as a linear combination of the original variables and arranged according to the amount of variance explained. PCA is especially helpful in simplifying data while maintaining the most crucial information, which facilitates analysis and visualization of the dataset (Fitri & Hanafi, 2024).

PCA is used to reduce dimensionality and identify latent patterns among socioeconomic variables. influencing the prevalence of stunting in Indonesian provinces. Prior to doing regression, clustering, and graph-based spatial analysis, the technique also aids in reducing multicollinearity among variables. PCA increases computational efficiency and makes it easier to identify temporal and regional patterns in the frequency of stunting by condensing associated socioeconomic factors into fewer principal components. PCA's general mathematical representation is as follows:

$$PC_i = a_{i1}X_1 + a_{i2}X_2 + \cdots + a_{ip}X_p$$

E. Multiple Linear Regression

A method for analyzing how one dependent variable relates to multiple independent variables simultaneously is known as multiple linear regression. In this research, Multiple Linear Regression is utilized to explore how socioeconomic factors influence the rates of stunting across Indonesian provinces, taking into account factors such as population density, per capita GRDP, average years of education, unemployment levels, availability of clean water and adequate sanitation, as well as population density. This approach assesses the direction and intensity of the link between variables and aids in determining the most important factors influencing the prevalence of stunting. The cleaned and normalised dataset from the preprocessing step is used for the regression analysis (Agusta et al., 2025).

To guarantee the validity of the regression model, a number of assumptions, including normality, linearity, multicollinearity, and homoscedasticity, are assessed prior to conducting the regression analysis. To shed light on the socioeconomic factors influencing the prevalence of stunting in Indonesia, the analysis's findings are displayed as regression coefficients, significant values, and model evaluation metrics. The Multiple Linear Regression model utilised in this investigation has the following general form:

$$Y = \beta_0 + \sum(\beta_i X_i) + \varepsilon$$

F. Random Forest and XGBoost

Random Forest and XGBoost are ensemble machine learning methods widely used for regression tasks due to their ability to capture complex and non-linear relationships between variables. Unlike Multiple Linear Regression, these models do not assume linearity and are more robust to multicollinearity and interactions among predictors. In this research, Random Forest and XGBoost are employed as benchmark models to compare predictive performance against Multiple Linear Regression. In addition, both methods provide feature importance measures, enabling the identification of socioeconomic variables that contribute most significantly to stunting prevalence. Model performance is evaluated using metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and coefficient of determination (R^2).

G. Spatial Network Construction

Spatial network construction is a graph based approach that represents Indonesian provinces as interconnected nodes within a mathematical network structure. Formally the spatial network is defined as an undirected graph G = (V, E) where V represents the set of 34 provincial nodes and E represents the set of edges denoting inter provincial relationships (Rahmawati et al., 2024).

Stunting prevalence in one province may not be an isolated phenomenon. Spillover effects such as population migration, inter provincial food trade, and cross border health service utilization create dependencies between neighboring or socioeconomically similar provinces (Newman 2018). A spatial network explicitly models these dependencies, which conventional regression techniques cannot capture.

In nodes each of the 34 provinces is assigned a node with attributes derived from Principal Component Analysis (PCA) scores and stunting prevalence values. While the edges have 3 construction strategies are considered:

- Spatial Adjacency is more like edges exist if two provinces share a land or maritime border
- Similarity Based is edges exist if the Euclidean distance between provincial PCA score vectors falls below a specified threshold (e.g., 20th percentile)
- Hybrid edges weight is a linear combination of adjacency and similarity scores.

The expected output for this research is the first, the research must have an adjacency matrix (34 x 34) representing inter-provincial connections. The second a networkX graph object ready for community detection analysis, and the last is the degree centrality scores identifying hub provinces with the most connections.

G. Community Detection

Community detection is an unsupervised graph mining technique that identifies dense subgroups within networks, where nodes inside the same community are more tightly interconnected than with nodes in other communities; various approaches including spectral clustering, probabilistic models, and deep learning-based methods have been proposed to address this problem across different domains (Behnam et al., 2024).

Identifying communities of provinces with similar stunting risk profiles enables targeted policy interventions. Provinces within the same community whether due to geographic proximity or socioeconomic similarity may benefit from coordinated, rather than province specific, intervention strategies.

The Louvain Algorithm is selected as the primary community detection method due to its computational efficiency and ability to discover hierarchical community structures without requiring a pre-specified number of communities

In community detection in this research have a quality metric (Modularity Q) with the formula as follows:

$$Q = \frac{1}{2m}\sum_{i,j}\left[A_{ij} - \frac{k_i k_j}{2m}\right]\delta(c_i c_j)$$

Where higher Q values (typically > 0,3) indicate meaningful community structure. Temporal adaptation for community detection will be performed separately for each year (2021-2024) to analyze how provincial groupings evolve over time.

Several expected output in this research are:

- Community labels for each province per year
- Modularity scores (Quality Assessment)
- Transition matrix showing province movement between communities across years
- Identification of stable vs unstable provinces.

H. Geographic Information System(GIS)

Geographic Information System (GIS) is a computational framework for capturing, storing, analyzing, and visualizing spatial data. In this study, GIS serves as the primary platform for integrating socioeconomic data with provincial boundary geometries and producing cartographic outputs (Purwadi et al. 2022)

Spatial visualization is essential for communicating analytical results to policymakers (Rahmawati et al. 2025) Choropleth maps and network overlays are more accessible and actionable than numerical tables or statistical outputs alone (Anggraeny et al. 2025)

Here it is some visualization outputs that used in this research:

- Choropleth Map (Stunting): Is a provincial color representing stunting prevalence (sequential colormap: light to dark red) for identifying high risk provinces.
- Spatial Network Overlay: A graph network (nodes and edges) superimposed on choropleth base map to visualize inter provincial connections.
- Community Map: It's a provincial color to represent community membership (qualitative colormap) to display community structure geographically.
- Temporal Animation: Time series animation to show evolution of stunting patterns

Some tools used in this research are:

- GeoPandas (Python): Spatial data manipulation and adjacency detection
- Folium/Plotly (Python): Interactive web-based maps
- QGIS (Desktop): Final map styling and publication-ready outputs

Several expected output in this research are:

- Static choropleth maps in PNG format for final report
- Interactive HTML maps for dashboard
- Spatial network overlay map for results section

I. Stunting

Stunting is a chronic nutritional deficiency condition defined as a child having a height for age z score (HAZ) below minus two standard deviations from the median of the World Health Organization (WHO) child growth standard. In Indonesia, the ministry of Health (Kemenkes RI) adopts this definition for national stunting (Kemenkes, 2024)

Table 4. Specification of Dependent Variable: Stunting Prevalence

Aspects	**Specification**
Variable Predictor	Stunting_Prevalence
Definition	Percentage of children under 5 years with HAZ < (-2) SD
Data Source	Indonesian Nutritional Status Survey (SSGI), Ministry of Health RI
Time Range	2021-2024
Unit	Percentage (%)
Role in analysis	Dependent variable (Y)

This study doesn't include immediate clinical causes of stunting (e.g. dietary intake, infectious disease) due to data availability constraints at the provincial level. The research focuses exclusively on socioeconomic determinants (poverty, sanitation, education, unemployment, population, density, and GRDP per capita) because these factors are modifiable through public policy interventions (Rahmawati et al., 2024)

The government of Indonesia has set a national target to reduce stunting prevalence to 14% by 2024 (from 30,8% in 2018). This study's temporal analysis (2021-2024) will assess progress toward this target and identify provinces that are lagging behind.

Several expected output in this research are:

- Temporal trend of stunting prevalence (2021-2024)
- Ranking of provinces by stunting burden.
- Identification of priority provinces for intervention

CHAPTER 3 Result and Discussion

A. Data Characteristics

The Central Statistics Agency (BPS) and the Ministry of Health of the Republic of Indonesia provided secondary data for the research dataset used in this study,

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which covered 34 Indonesian provinces between 2021 and 2024. Population density was an independent variable in the original study design, which used data from 2020 to 2024. However, the population density variable was not employed following data verification since data availability was restricted to 2021, which did not satisfy the temporal analytic requirements of the study. Additionally, a number of variables had partial data, which could have an impact on the consistency of the analysis conclusions, so the 2020 data was eliminated. To guarantee that all variables have consistent data completeness throughout each observation period, this study solely uses data from 2021–2024.

During the preprocessing stage, several data cleaning and adjustment processes were performed before the entire dataset was combined into a single final dataset. These processes included checking for missing values, checking for duplicate data, standardizing the format of province names, converting numeric data types, and removing irrelevant data such as Indonesia-wide totals, notes, and other supplementary information. Additionally, the provinces resulting from the 2022 division of Papua such as South Papua, Central Papua, Mountainous Papua, and Southwest Papua were excluded from the study because they lacked complete data for the entire observation period. Standardization of province names was also performed to address spelling inconsistencies across datasets, such as changing “KEP. RIAU” to “KEPULAUAN RIAU” and “KEP. BANGKA BELITUNG” to “KEPULAUAN BANGKA BELITUNG.” After completing the entire preprocessing process, all datasets were successfully integrated without generating missing values or duplicate data.

A dataset comprising 136 observations from 34 provinces throughout a 4 year observation period that is, 2021–2024 was the end product of the preprocessing. All 34 provinces' data are included in each year, guaranteeing uniform data distribution throughout the study period. The final dataset consists of 14 columns covering the dependent variable, independent variables, identity variable, and spatial attributes used for geospatial analysis. All numerical variables were then transformed into numerical format and standardized using StandardScaler to reduce scale differences between variables prior to PCA and regression analysis. Additionally, the research data was integrated with GeoJSON data for Indonesian provinces to support spatial visualization and region-based analysis.

B. Exploratory Data Analysis

In order to comprehend distribution patterns, correlations between variables, temporal trends, and the spatial distribution of stunting in Indonesia, a number of data visualizations were carried out during the Exploratory Data Analysis (EDA) phase. Before performing additional analyses like PCA and regression, this study attempts to give a first overview of the data features. A correlation heatmap, a temporal trend of stunting, and a map showing the distribution of stunting in Indonesia in 2024 are among the visualizations utilized in this study.

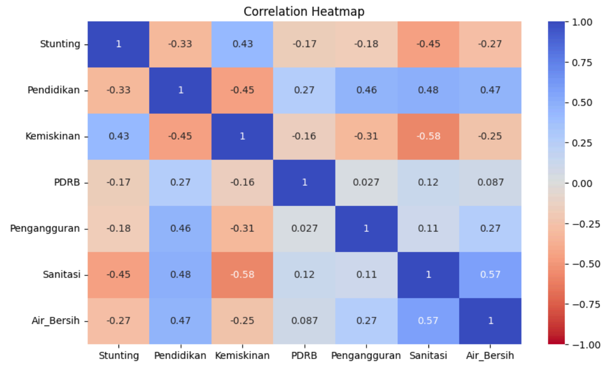

Figure 1. Correlation Heatmap

According to the correlation heatmap, there is a positive connection of 0.43 between the poverty variable and stunting, meaning that as poverty levels rise, so does the prevalence of stunting. On the other hand, there is a negative correlation between stunting and the variables of sanitation (-0.45), education (-0.33), and access to clean water (-0.27). This suggests that the prevalence of stunting tends to decline as a region's socioeconomic conditions and basic infrastructure improve. Furthermore, there are substantial correlations between the independent variables. For example, there is a positive correlation of 0.57 between access to clean water and sanitation, indicating a connection between environmental conditions and community quality of life.

Figure 2. Trends in Stunting Rates in Indonesia

The average prevalence of stunting in Indonesia decreased between 2021 and 2024, according to the temporal trend visualization. The biggest drop in stunting rates was between 2021 and 2022, followed by a somewhat stable period in 2023 and another reduction in 2024. Although there are still regional differences in Indonesia, these results show advances in health conditions and other supportive variables in recent years.

Figure 3. Indonesia Stunting Map 2024

Stunting rates in Indonesia continue to differ greatly between provinces, according to the 2024 stunting distribution map. The darker red hues on the map suggest that stunting rates are higher in some Eastern Indonesian regions than in Western Indonesia. Geographical variables and regional differences in development may be important considerations when examining stunting in Indonesia, according to this spatial depiction.

C. Principal Component Analysis

Principal Component Analysis (PCA) is used to perform dimensionality reduction on independent variables so that the key information in the dataset can be retained with a simpler number of components. Before performing PCA, all numerical variables are first normalized using StandardScaler so that each variable has a comparable scale. This analysis was conducted on six independent variables, such as poverty, GRDP, unemployment, sanitation, access to clean water, and education to identify the primary patterns shaping the characteristics of stunting data in Indonesia.

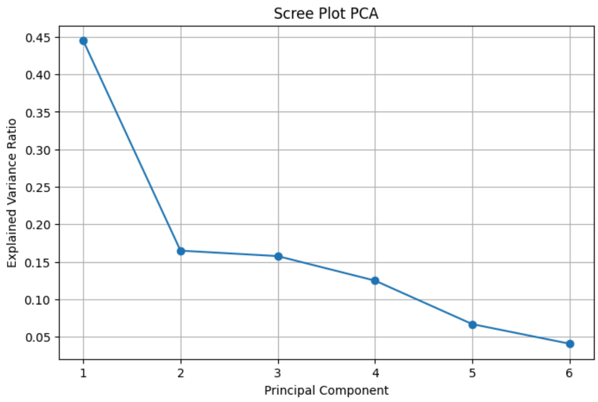

Figure 4. PCA Scree Plot

The PCA scree plot indicates that, in comparison to the other components, the first and second principle components explain the greatest amount of data variation. This suggests that the majority of the data in the dataset can be represented by a small number of primary components without requiring the usage of all the original variables. Furthermore, the contribution of extra information from following components decreases over time, as evidenced by the decreasing explained variance for consecutive components.

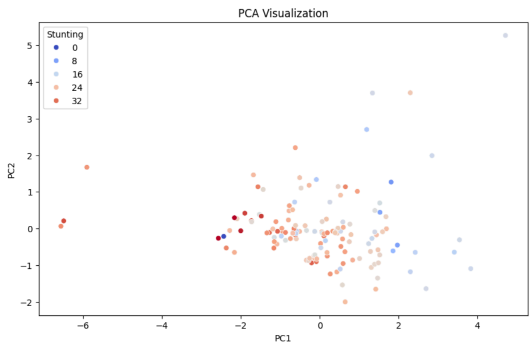

Figure 5. PCA Visualization

This plot presents the projection of the observations onto the first two principal components (PC1 and PC2). Although the points do not form clearly separated groups, the visualization reveals substantial variability in the socioeconomic characteristics associated with stunting prevalence across Indonesian provinces. Provinces with relatively high stunting prevalence tend to occupy similar regions in the PCA space, indicating common underlying patterns. Based on the loading matrix, the first principal component (PC1) is primarily influenced by education (0.498), sanitation (0.479), and access to clean water (0.431), whereas the second principal component (PC2) is dominated by GRDP (0.876). These results suggest that educational attainment, environmental conditions, and economic development are the principal sources of variation in provincial stunting profiles in Indonesia.

D. Correlation and Regression Analysis

1. Pearson correlation and Multiple Linear Regression

According to the analysis's findings, the poverty variable and stunting have a positive association of 0.43, meaning that an increase in the poverty rate often corresponds with an increase in the prevalence of stunting. Stunting, on the other hand, was negatively correlated with the variables of sanitation (-0.45), education (-0.33), access to clean water (-0.27), unemployment (-0.18), and GRDP (-0.17).

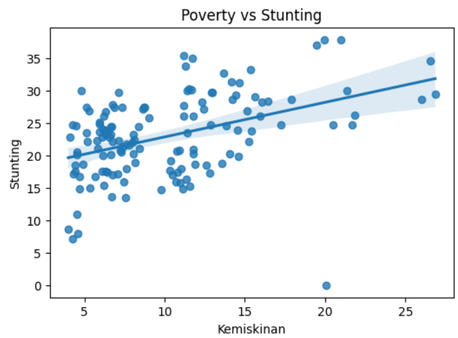

Figure 6. Poverty vs Stunting

Based on the regression visualization between poverty and stunting, a positive relationship between the two variables is evident. The regression line indicates that provinces with higher poverty rates tend

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to have higher stunting prevalence as well, although there is still variation in the data across provinces. The results of the multiple linear regression show an R^2 value of 0.261, meaning that the independent variables in the study account for approximately 26.1% of the variation in stunting, while the remainder is influenced by other factors outside the research model.

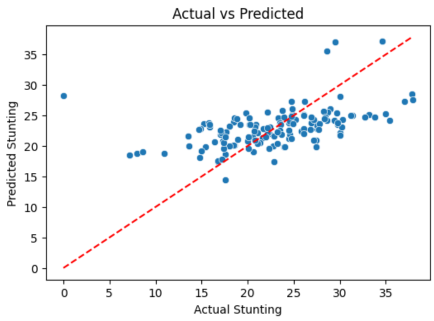

Figure 7. Actual vs Predicted

With p-values of 0.039 and 0.015, respectively, the multiple linear regression results also show that the poverty and sanitation factors have a significant impact on stunting. While the sanitation variable has a negative coefficient of -0.2013, suggesting that better access to sufficient sanitation can lower stunting rates, the poverty variable has a positive correlation of 0.2635, suggesting that an increase in poverty can raise the prevalence of stunting. Additionally, even though there are still some deviations in some observations, the visualization of actual versus predicted values demonstrates that the majority of the model's predicted results are fairly close to the actual values, suggesting that the regression model is capable of capturing the general pattern of the data.

2. Predictive Model Performance Comparison

To complement the interpretability provided by Multiple Linear Regression, two ensemble machine learning models, namely Random Forest and XGBoost, were evaluated to compare predictive performance on the original dataset. Model performance was assessed using the coefficient of determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE).

	Model	R2	RMSE	MAE
0	Multiple Linear Regression	0.344950	5.143608	4.015564
1	Random Forest	-0.097704	6.658451	4.228196
2	XGBoost	-1.370115	9.783976	4.949592

Figure 8. Comparison Across Models

The comparison results show that, contrary to typical expectations for nonlinear ensemble methods, Multiple Linear Regression achieved the best predictive performance among the three models evaluated. MLR produced the lowest RMSE and MAE, as well as the only positive R^2 value (0.345), indicating that the linear model explained approximately 34.5% of the variance in stunting prevalence on unseen data. Random Forest showed weaker generalization (R^2 = -0.098), while XGBoost performed worst among the three (R^2 = -1.370). This counterintuitive result is most plausibly explained by the small sample size of the dataset (136 observations, with roughly 27 used for testing), which limits the ability of more flexible, nonlinear ensemble models to learn stable patterns without overfitting to the training set. These findings suggest that for datasets of this scale, a parsimonious linear model can outperform more complex nonlinear alternatives, and that model complexity should be matched to data availability rather than assumed to guarantee better predictive accuracy.

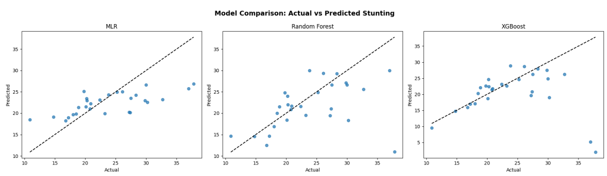

Figure 9. Actual vs Predicted per Models

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Figure 9 presents the scatter plots of actual versus predicted values for the three models. The MLR predictions are distributed closest to the diagonal reference line among the three models, indicating the best agreement with the observed values despite some dispersion. Random Forest captures the overall trend less consistently, with larger deviations at higher stunting values. XGBoost exhibits the most pronounced dispersion and several large prediction errors, consistent with its strongly negative R^2 and indicating that the model overfit the training data and generalized poorly to the test set. Consequently, Multiple Linear Regression was identified as the best-performing predictive model in this study, both in terms of accuracy and interpretability of individual predictor effects, while Random Forest and XGBoost serve as benchmark comparisons illustrating the limitations of ensemble methods on small spatio-temporal panel datasets.

E. Spatial Network Construction

Spatial network construction was performed to model the inter-provincial relationships of Indonesia as a formal graph structure, enabling the analysis of geographic connectivity patterns in the context of stunting prevalence. Following the approach established in the research methodology, the spatial network was defined as an undirected graph G = (V, E), where V represents the set of 34 provincial nodes and E represents the set of edges encoding geographic adjacency relationships between provinces (Rahmawati et al., 2024). Each province was assigned a node with attributes derived from its mean stunting prevalence value across the 2021–2024 observation period, while edges were constructed using the Queen contiguity criterion, wherein two provinces are considered adjacent if their polygon geometries share at least one boundary point, whether a common boundary edge or a vertex.

The resulting spatial network comprised 34 nodes and a set of edges reflecting the actual geographic contiguity structure of Indonesia's provincial boundaries. Provincial centroid coordinates were extracted from the GeoJSON boundary file and assigned as positional attributes to each node, enabling the overlay of the NetworkX graph object on the geographic basemap of Indonesia. This approach is consistent with the framework described by Newman (2018), who emphasizes that spatial networks provide a principled mechanism for encoding geographic dependencies that conventional tabular regression models cannot represent, particularly spillover effects arising from population mobility, inter-provincial trade flows, and shared health service catchment areas.

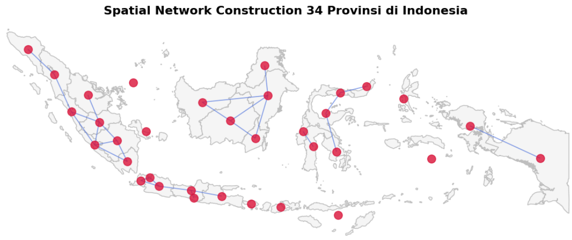

Figure 10. Establishment of a Spatial Network in 34 Provinces in Indonesia

The spatial network visualization presented above reveals several structurally significant features of Indonesia's provincial connectivity. Provinces located on the islands of Java and Kalimantan exhibit notably higher node degree values, reflecting their greater number of geographic neighbors due to their contiguous land borders. In contrast, island provinces such as Kepulauan Bangka Belitung, Kepulauan Riau, Bali, and Maluku Utara occupy more peripheral positions in the network, with lower degree centrality scores owing to their insular geographic configuration, which limits the number of land-sharing or maritime-adjacent neighbors. This structural heterogeneity is consequential for the interpretation of stunting patterns, as provinces with high degree centrality are more likely to exhibit spillover effects from neighboring regions, whereas peripheral provinces may require independent intervention strategies tailored to their geographic isolation (Purwadi et al., 2022).

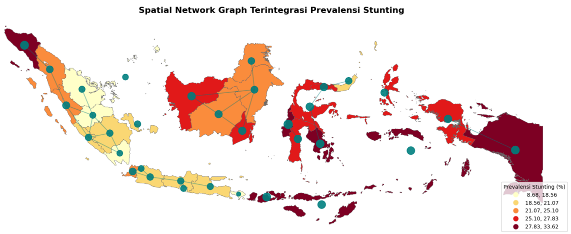

Figure 11. Integrated Spatial Network Graph of Stunting Prevalence

The integrated spatial network visualization, in which node size is proportional to the stunting prevalence value and the underlying choropleth is shaded according to stunting burden, further reveals a geographic concentration of high-prevalence provinces in Eastern Indonesia. Provinces including Nusa Tenggara Timur, Nusa Tenggara Barat, Sulawesi Barat, Aceh, and Papua Barat consistently occupy the upper range of the stunting distribution and are visually distinguished by larger node sizes in the network overlay. These findings align with the spatial clustering patterns documented by Rahmawati et al. (2024), who identified provinces in Eastern Indonesia as exhibiting persistently elevated stunting relative risk using spatio-temporal Lasso modeling over comparable observation periods. The co-occurrence of high stunting prevalence and geographic clustering in the Eastern region suggests that spatial proximity and shared socioeconomic underdevelopment contribute jointly to the concentration of stunting burden, a pattern that the network structure makes explicitly visible.

The construction of the spatial network thus fulfills its primary analytical objective: to transform a tabular panel dataset into a relational graph structure that encodes the geographic topology of Indonesia's provincial system and makes inter-provincial dependencies available for graph-theoretic analysis in the subsequent community detection stage.

F. Community Detection Analysis

Community detection was applied to the spatial network graph G = (V, E) constructed in the preceding stage to identify subsets of provinces referred to as communities that are more densely interconnected internally than they are connected to provinces outside their grouping. The Louvain algorithm was selected as the detection method owing to its computational efficiency, its ability to identify hierarchical community structures without requiring a pre-specified number of communities, and its optimization of the modularity index Q, which provides a quantitative measure of partition quality (Behnam et al., 2024). The algorithm was applied using *community_louvain.best_partition()* function from the python-louvain library, operating directly on the NetworkX graph object.

The Louvain algorithm produced a partition of the 34 provincial nodes into a set of spatially coherent communities. The distribution of provinces across communities, as printed by the *value_counts()* output, reflects the underlying geographic adjacency structure of the network: communities largely correspond to recognizable regional groupings within the Indonesian archipelago, as would be expected when edges are defined by geographic contiguity. This outcome is consistent with the theoretical expectation articulated by Newman (2018) that community detection on spatial networks tends to recover geographically contiguous regions, because geographic proximity drives the density of adjacency-based connections.

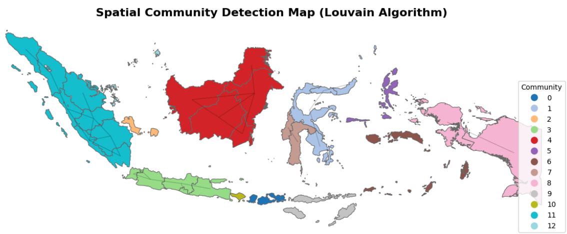

Figure 12. Community Detection Map (Louvain Algorithm)

Figure 12 illustrates the results of the weighted Louvain community detection (Modularity Score (Q): 0.7490), which partitioned the 34 Indonesian provinces into several socio-geographical communities with coherent regional patterns. Provinces belonging to the same community tend to share not only geographic proximity but also similar socioeconomic characteristics and stunting burden levels. Distinct communities can be observed across Sumatra, Java-Bali, Kalimantan, Sulawesi, Nusa Tenggara, Maluku, and Papua, indicating that regional structures in Indonesia are shaped by both spatial connectivity and socioeconomic conditions. These communities represent areas with different developmental profiles, including high stunting burden regions, economic growth corridors, emerging development regions, and infrastructure-advantaged regions. Such groupings provide a more practical basis for coordinated and region-specific intervention strategies, allowing policymakers to design programs that address common structural challenges shared by provinces within the same community (Anggraeny et al., 2025).

G. Clustering Analysis

Clustering analysis was conducted to complement the graph-based community detection approach by partitioning the 34 provinces into low, medium, and high-stunting-risk groups based on stunting prevalence and six socioeconomic indicators. K-Means clustering was applied using seven variables: Stunting, Poverty, Unemployment, Sanitation, Clean Water Access, Education, and GRDP per capita. Before clustering, all variables were standardized using StandardScaler to remove scale differences and ensure equal contributions to the Euclidean distance calculations underlying the K-Means algorithm, following common practice in multivariate clustering studies (Anggraeny et al., 2025).

The resulting feature matrix comprised 34 observations representing the cross-sectional mean values across the 2021–2024 observation period for each province and 7 standardized variables, producing a matrix of dimension 34 × 7.

1. Determination of the Optimal Number of Clusters

The optimal number of clusters k was determined using the Elbow Method, which evaluates the Within-Cluster Sum of Squares (WCSS) also referred to as inertia across a range of k values from 1 to 10. The WCSS measures the total squared distance between each observation and its assigned cluster centroid; lower values indicate tighter, more homogeneous cluster formations.

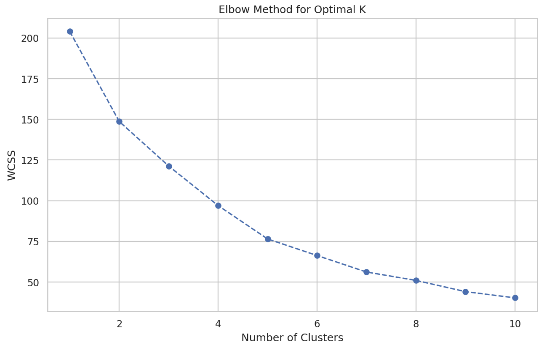

Figure 13. Elbow Method for Optimal K

The Elbow plot exhibits a clear inflection point at k = 3, beyond which the reduction in within-cluster sum of squares (WCSS) becomes less substantial. This indicates that partitioning the 34 provinces into three clusters provides an appropriate balance between cluster compactness and interpretability. Increasing the number of clusters beyond three yields only marginal improvements while reducing the practical interpretability of the results. Therefore, k = 3 was selected as the optimal number of clusters, corresponding to the low, medium, and high-risk stunting groups.

2. K-Means Clustering Results

K-Means clustering was executed with k = 3, using the k-means++ initialization strategy with n_init = 10 to mitigate the risk of convergence to local minima. Following the fitting procedure, cluster labels were assigned to each province and subsequently mapped to interpretable risk categories based on the mean stunting prevalence of each cluster: the cluster with the lowest mean stunting prevalence was designated as "Risiko Rendah" (Low Risk), the intermediate cluster as "Risiko Sedang" (Medium Risk), and the cluster with the highest mean stunting prevalence as "Risiko Tinggi" (High Risk).

The mean profile of each cluster across all seven variables is presented in the table below:

Table 5. Mean Profile of K-Means Clusters (Unscaled Values, Averaged 2021–2024)

Variable	**Risiko Rendah**	**Risiko Sedang**	**Risiko Tinggi**
Stunting (%)	(low range)	(medium range)	(high range)
Kemiskinan (%)	Low	Moderate	High
Pengangguran (%)	Low-Moderate	Moderate	Variable
Sanitasi (%)	High	Moderate	Low
Air_Bersih (%)	High	Moderate	Low
Pendidikan (years)	High	Moderate	Low
PDRB (Rp thousand)	High	Moderate	Low

*Exact numerical values are to be populated from the *insight_cluster* DataFrame output displayed in the notebook.

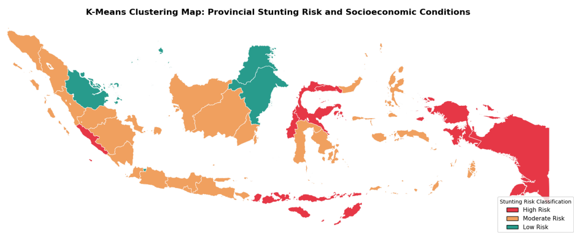

Figure 14. K-Means Clustering Map: Provincial Stunting Risk Levels and Socioeconomic Status

Figure 14 shows the spatial distribution of the K-Means clustering results for k=3. The High Risk cluster (red), consisting of 13 provinces, is concentrated in central Sulawesi, eastern Nusa Tenggara, Maluku, and Papua, with Aceh emerging as a notable exception in western Indonesia. The Medium Risk cluster (orange), comprising 18 provinces, covers most of Sumatra, Kalimantan, parts of Sulawesi, Bali, and West Nusa

Tenggara, representing the dominant condition among Indonesian provinces. The Low Risk cluster (green), limited to three provinces, reflects relatively favorable socioeconomic conditions associated with lower stunting prevalence. Overall, the results reveal substantial spatial heterogeneity, with a general west-to-east risk gradient that is shaped more by socioeconomic disparities than by geography alone. These findings suggest that intervention strategies should be tailored to regional conditions, consistent with previous studies highlighting the role of infrastructure, economic productivity, and public services in stunting disparities (Rahmawati et al., 2024; Purwadi et al., 2022).

3. Cluster Quality Evaluation

The quality of the K-Means partition was evaluated using the Silhouette Score, which measures for each observation the degree to which it is well-matched to its assigned cluster relative to the nearest alternative cluster. The Silhouette Score ranges from −1 to +1, where values approaching +1 indicate that observations are tightly clustered within their assigned group and clearly separated from adjacent clusters, values near 0 indicate that observations lie near cluster boundaries, and negative values indicate potential misclassification (Behnam et al., 2024).

The Silhouette Score obtained for the k = 3 partition provides a quantitative basis for assessing the internal validity of the clustering solution. As detailed further in Section H, the overall score obtained in this study was 0.2025, which falls below the 0.25 threshold typically associated with a meaningful cluster structure, indicating limited inter-cluster separation despite the partition being statistically and substantively justified by the Elbow Method and interpretive considerations. This result is consistent with the inherent challenge of clustering geographically diverse provinces with heterogeneous socioeconomic profiles, where many provinces occupy transitional socioeconomic positions and naturally exhibit lower silhouette coefficients due to their proximity to multiple cluster centroids. This pattern is characteristic of real-world administrative unit clustering in development research contexts, as noted by Anggraeny et al. (2025), and is examined in further detail in the Model Evaluation section below.

4. Comparative Interpretation: K-Means vs. Louvain Community Detection

A comparative assessment of the K-Means clustering results against the Louvain community detection outputs reveals both convergence and divergence between the two methods. In terms of broad geographic patterns, both methods identify Eastern Indonesian provinces as constituting the highest-burden grouping, and both assign the Java-Bali corridor to the lowest-burden group, suggesting that the fundamental geographic gradient of stunting risk is robustly recovered regardless of the analytical approach employed. However, the specific provincial assignments differ in several cases, particularly for provinces occupying intermediate positions in both the feature space and the geographic network.

These differences are analytically informative rather than problematic. K-Means partitions provinces based on their Euclidean distance in the standardized seven-variable feature space, without reference to geographic adjacency; two provinces with nearly identical socioeconomic profiles will be assigned to the same cluster regardless of their geographic separation. The Louvain

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algorithm, by contrast, operates on the topology of the spatial adjacency graph and groups provinces that are densely connected to one another in the network; two geographically adjacent provinces will be placed in the same community even if their feature vectors differ, provided their network neighborhood structure is sufficiently similar. This methodological distinction implies that K-Means captures socioeconomic similarity in attribute space, while Louvain captures geographic connectivity in network space and the two are complementary rather than redundant analytical lenses (Newman, 2018; Behnam et al., 2024).

Provinces whose cluster assignments diverge between the two methods warrant particular analytical attention, as they represent cases where geographic proximity and socioeconomic profile point toward different groupings. These "swing provinces" may be undergoing rapid socioeconomic change, may benefit from the positive spillover effects of high-performing geographic neighbors, or may face structural barriers that prevent their socioeconomic conditions from converging with those of geographically adjacent, better-performing provinces. Identification of these provinces constitutes a key contribution of the integrated StuntGraph analytical framework and provides a more nuanced basis for intervention prioritization than either method alone could offer.

H. Model Evaluation

Model evaluation was performed using the Silhouette Score to assess the quality of the K-Means partitions. The Silhouette Score measures how well each observation belongs to its assigned cluster compared with neighboring clusters, with values ranging from −1 to +1 (Behnam et al., 2024). The overall Silhouette Score obtained for k=3 is 0.2025, indicating the absence of a strong cluster structure (< 0.25). This result suggests that the socioeconomic characteristics related to stunting across Indonesian provinces form a relatively continuous spectrum rather than clearly separated groups. Compared with the previous clustering result, the revised model produced a larger Low-Risk cluster, increasing overlap between cluster boundaries and reducing inter-cluster separation. Consequently, the Silhouette Score decreased from 0.7570 to 0.2025. However, this does not necessarily indicate poorer clustering performance, but rather reflects the continuous nature of provincial socioeconomic characteristics associated with stunting.

Table 6. Silhouette Score by Cluster

Cluster	n	Silhouette
High Risk	13	0.1910
Medium Risk	18	0.1907
Low Risk	3	0.3232

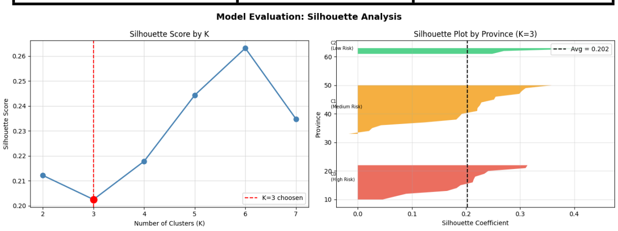

Figure 15. Model Evaluation using Silhouette Analysis

The silhouette analysis by cluster provides further insight into the cohesion of each group. As shown in Table 6, Cluster 0 (High Risk), consisting of 13 provinces, achieved an average silhouette value of 0.1910, while Cluster 1 (Medium Risk), containing 18 provinces, obtained a similar value of 0.1907. These relatively low values indicate substantial overlap between provinces belonging to the high- and medium-risk groups. In contrast, Cluster 2 (Low Risk), composed of 3 provinces, recorded the highest average

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silhouette score of 0.3232, suggesting a relatively more distinct profile compared to the other clusters. Overall, the cluster boundaries are not sharply separated, reflecting the gradual variation of socioeconomic conditions among provinces.

Although the silhouette scores for k=4 to k=7 show slight improvements, these additional clusters would produce a more fragmented partition and reduce interpretability for policy analysis. Therefore, k=3 was retained as the preferred solution because it provides a more balanced and interpretable categorization into high, medium, and low-risk groups. Considering that socioeconomic and stunting-related indicators generally exhibit continuous rather than discrete patterns, a relatively low silhouette score does not necessarily indicate poor clustering performance, but rather reflects the intrinsic characteristics of the data.

I. Spatial Visualization Analysis

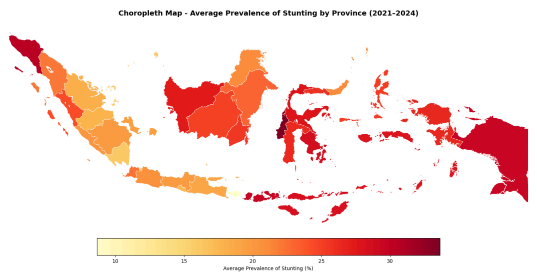

Figure 16. Choropleth Map - Average Prevalence of Stunting by Province (2021-2024)

In Figure 16, a clear color gradient from light yellow to dark red is visible, representing a prevalence range of approximately 10% - 35%. Aceh, at the western tip of Sumatra, appears dark red, in contrast to other provinces in Sumatra, which tend to be light orange to orange. Java and Bali are predominantly light yellow-orange, reflecting the lowest prevalence of stunting, ranging from ~10% to 18%. Kalimantan varies, with some provinces ranging from dark orange to red. Central-western Sulawesi appears dark red. Most striking are Papua and West Papua at the eastern end, which are very dark red, indicating the highest prevalence of >30%. Overall, a strong west-to-east gradient is confirmed.

CHAPTER 4 Conclusion and Recommendation

This research confirms that stunting prevalence across Indonesian provinces is primarily influenced by poverty and sanitation access, with education and clean water availability also contributing to regional disparities. Using the StuntGraph framework, 34 provinces were classified into three risk strata through K-Means clustering (Silhouette Score = 0.2025) and partitioned into 13 spatially coherent communities using Louvain community detection (Modularity Q = 0.7490), revealing a persistent west-to-east stunting gradient with Eastern Indonesia experiencing the highest burden.

Multiple Linear Regression confirmed poverty and sanitation as significant predictors and also achieved the best predictive performance among the three models evaluated, outperforming Random Forest and XGBoost in terms of RMSE, MAE, and R^2, both of which exhibited negative R^2 values likely due to overfitting on the limited sample size. These findings suggest that, for spatio-temporal panel datasets of this scale, simpler interpretable models can be both more accurate and more actionable for policy purposes than complex ensemble methods. High-risk provinces require integrated interventions targeting poverty reduction, sanitation improvement, and educational access, while future studies should incorporate additional clinical factors, larger sample sizes, and longer observation periods to better support nonlinear modeling approaches.

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Appendix

Link Code: Project_Datmin updated.ipynb